

SHYAM SARAN P

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SUMMARY

Final-year IT student experienced in building full-stack web applications and AI-integrated systems. Skilled in backend development, RAG-based document search, and machine learning with Python.

EDUCATION

SSN College of Engineering, Chennai	Expected May 2027
B.Tech in Information Technology	CGPA: 7.9
Doveton Boys Higher Secondary School, Chennai	2023
Higher Secondary (Class XII)	90.2%

EXPERIENCE

Junior AI Analyst (Part-Time) A2000 Solutions	Jun 2026 – Present
- Developed an educational mentor AI assistant by fine-tuning a Small Language Model (SLM) using QLoRA. - Created synthetic JSONL training datasets and prompt templates to improve the assistant's response quality.	
Software Development Intern Highways Department, Government of Tamil Nadu	May 2026 – Jun 2026
- Developed a full-stack portal using React, FastAPI, and PostgreSQL to streamline highway tender and administrative workflows. - Implemented secure JWT authentication, hierarchical Role-Based Access Control (RBAC), and detailed audit logs.	
Web Developer Intern A2000 Solutions	Jun 2025 – Jul 2025
- Built a complete Learning Management System (LMS) using Flask and PostgreSQL with role-based access for students, instructors, and administrators. - Developed REST APIs for course management and learning workflows, optimizing SQL queries to improve application performance.	

PROJECTS

VeriSource AI Python, FastAPI, PostgreSQL, ChromaDB, ONNX, React	GitHub
- Built a RAG-based platform for educational institutions to search rules and regulations documents and find relevant information. - Implemented document search using ChromaDB and ONNX Runtime, reducing memory usage by 60%.	
SDGSync – Enterprise Project Alignment Platform Python, FastAPI, PostgreSQL, React	GitHub
- Built a web platform to categorize project proposals based on UN Sustainable Development Goals (SDGs). - Used a BERT-based classifier with a RAG pipeline, achieving 95% classification accuracy.	
AI-Powered Examination Portal (Ongoing) FastAPI, React.js, PostgreSQL	GitHub
- Developed an examination portal with automated MCQ generation and assessment using FastAPI, React, and PostgreSQL. - Currently building a PyTorch-based proctoring engine to analyze facial expressions, gaze direction, head pose, and audio signals for real-time behavioral monitoring.	

SKILLS

Languages	Python, Java (Intermediate), C
Frameworks & APIs	FastAPI, Flask, React (Basic), RESTful APIs
AI Integration	Retrieval-Augmented Generation (RAG)
Databases & Tools	PostgreSQL, MySQL, SQLite, Git, GitHub, VS Code
Concepts	Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PUBLICATIONS

A Safety-Certified Framework for Maternal Triage: Integrating Conformal Prediction and Cost-Sensitive Objective Optimization	
IEEE ICCVBIC 2026, Lalitpur, Nepal	Third Author DOI
Reddit Rule-Violation Classification Using LLM-Assisted Data Augmentation	
IEEE ICDSAAI 2026	Third Author DOI
Explainable and Robust Heritage Intelligence Framework (ERHIF)	
International Journal of Advanced Science and Engineering (IJASE), 2026	Fourth Author DOI